PRAKTIKUM BASIS DATA

MODUL 09

Scripting Procedure dan Function

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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Jurnal

1. create table employees as select * from hremployees.

Query:

```
--

create table employees as select * from hremployees;
```

Analisis:

Membuat tabel baru employees dengan menyalin seluruh struktur kolom dan data dari tabel hremployees. yang disalin hanya kolom dan data, sedangkan constraint seperti primary key, foreign key, dan index tidak ikut tersalin.

2. select * from employees.

Query:

```
select * from employees;
```

Output:

Script Output × | Query Result ×

SQL | All Rows Fetched: 10 in 0.019 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-09-2005	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-01-2001	AD_VP	17000	(null)	100	90
4	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-01-2006	IT_PROG	9000	(null)	102	60
5	104	Bruce	Ernst	BERNST	590.423.4568	21-05-2007	IT_PROG	6000	(null)	103	60
6	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-02-2007	IT_PROG	4200	(null)	103	60
7	124	Kevin	Mourgos	KMOURGOS	650.123.5234	16-11-2007	ST_MAN	5800	(null)	100	50
8	141	Trenna	Rajs	TRAJS	650.121.8009	17-10-2003	ST_CLERK	3500	(null)	124	50
9	149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-01-2008	SA_MAN	10500	0,2	100	80
10	174	Ellen	Abel	EABEL	011.44.1644.429267	11-05-2004	SA_REP	11000	0,3	149	80

Analisis:

Data yang ditampilkan di employees sama persis dengan data di hremployees saat proses copy dilakukan.

3.

```
CREATE OR REPLACE PROCEDURE increase_salary (
department_id_in   IN employees.department_id%TYPE,
    increase_pct_in    IN NUMBER)
     IS
BEGIN
    FOR employee_rec
            IN (SELECT employee_id
            FROM employees
           WHERE department_id = increase_salary.department_id_in)
    LOOP
            UPDATE employees emp
                SET emp.salary = emp.salary + emp.salary * increase_salary.increase_pct_in
            WHERE emp.employee_id = employee_rec.employee_id;
            DBMS_OUTPUT.PUT_LINE ('Updated ' || SQL%ROWCOUNT);
    END LOOP;
END increase_salary;
```

Query:

```
CREATE OR REPLACE PROCEDURE increase_salary (
    department_id_in IN employees.department_id%TYPE,
    increase_pct_in  IN NUMBER
)
IS
BEGIN
    FOR employee_rec
        IN (SELECT employee_id
            FROM employees
            WHERE department_id = increase_salary.department_id_in)
    LOOP
        UPDATE employees emp
        SET emp.salary = emp.salary + emp.salary * increase_salary.increase_pct_in
        WHERE emp.employee_id = employee_rec.employee_id;

        DBMS_OUTPUT.PUT_LINE('Updated ' || SQL%ROWCOUNT);
    END LOOP;
END increase_salary;
```

Analisis:

Stored procedure bernama increase_salary yang digunakan untuk menaikkan gaji karyawan berdasarkan department tertentu.

4. BEGIN increase_salary (90, .50); ROLLBACK; END; / select * from employees where department_id = 90

Query:

```
BEGIN
    increase_salary (90, .50);
    ROLLBACK;
END;
/
select * from employees where department_id = 90
```

Output:

Script Output × | Query Result ×

SQL | All Rows Fetched: 3 in 0.009 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-09-2005	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-01-2001	AD_VP	17000	(null)	100	90

Analisis:

Menaikkan gaji karyawan di departemen 90 sebesar 50%, namun perubahan tersebut langsung dibatalkan dengan rollback.

5. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).

Query:

```
BEGIN
    increase_salary (90, .50);
END;
/
select * from employees where department_id = 90
```

Output:

Script Output × | Query Result ×

SQL | All Rows Fetched: 3 in 0.004 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-06-2003	AD_PRES	36000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-09-2005	AD_VP	25500	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-01-2001	AD_VP	25500	(null)	100	90

Analisis:

Perubahan yang dilakukan oleh procedure increase_salary akan tetap berlaku dan terlihat pada hasil SELECT, yaitu gaji karyawan di departemen 90 meningkat 50%.